

EXP NO :1 Setting up Hyperledger Fabric Development Environment

Aim:

To install and configure the Hyperledger Fabric development environment on Linux, create the Fabric test network, and create an application channel named "mychannel".

Requirements

- Linux Operating System (Kali)
- Docker and Docker Compose
- Go Programming Language
- Git and Curl
- Hyperledger Fabric Samples, Binaries and Docker Images

Procedure

1. Update the system

Command(s):

- `sudo apt update`
- `sudo apt upgrade -y`

2. Verify Docker installation

Command(s):

- `docker --version`
- `sudo systemctl start docker`
- `sudo systemctl enable docker`
- `docker run hello-world`

```
(mohana@kali)-[~]  
$ docker --version  
Docker version 28.5.2+dfsg4, build 9cc6dea35e9a963f281434761c656fba4ac43aed
```

```
(mohana@kali)-[~]  
$ sudo systemctl start docker  
  
(mohana@kali)-[~]  
$ sudo systemctl enable docker  
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
```

```
(mohana@kali)-[~]  
$ docker run hello-world  
  
Hello from Docker!  
This message shows that your installation appears to be working correctly.  
  
To generate this message, Docker took the following steps:  
1. The Docker client contacted the Docker daemon.  
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.  
   (amd64)  
3. The Docker daemon created a new container from that image which runs the  
   executable that produces the output you are currently reading.  
4. The Docker daemon streamed that output to the Docker client, which sent it  
   to your terminal.  
  
To try something more ambitious, you can run an Ubuntu container with:  
$ docker run -it ubuntu bash  
  
We are having trouble restoring your session.  
Share images, automate workflows, and more with a free Docker ID:  
https://hub.docker.com/  
For more examples and ideas, visit:  
https://docs.docker.com/get-started/
```

3. Verify Go installation

Command(s):

- go version

```
(mohana@kali)-[~]  
$ go version  
go version go1.26.5 linux/amd64
```

4. Create Fabric workspace

Command(s):

- mkdir -p ~/fabric-dev
- cd ~/fabric-dev

5. Download Hyperledger Fabric

Command(s):

- curl -sSL

<https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh> -o install-fabric.sh

- chmod +x install-fabric.sh
- ./install-fabric.sh docker samples binary

```
(mohana@kali)~[~/fabric-dev]
$ curl -sSL https://bit.ly/2ysb0FE | bash -s

Clone hyperledger/fabric-samples repo

=> Changing directory to fabric-samples
fabric-samples v2.5.16 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric binaries

=> Downloading version 2.5.16 platform specific fabric binaries
=> Downloading: https://github.com/hyperledger/fabric/releases/download/v2.5.16/hyperledger-fabric-linux-amd64-2.5.16.tar.gz
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left     Speed
  0     0    0     0    0     0      0      0      0     0  0:00:00  0:00:00  0:00:00     0
12 121.8M 12 15.34M    0    153.7k    0   13:31   01:42   11:49 238.1kZ
zsh: done      curl -sSL https://bit.ly/2ysb0FE |
zsh: suspended bash -s
```

6. Configure PATH

Command(s):

- echo 'export PATH=\$PATH:\$HOME/fabric-dev/fabric-samples/bin' >> ~/.zshrc
- source ~/.zshrc

```
(mohana@kali)~[~/fabric-dev/fabric-samples/test-network]
$ echo 'export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin' >> ~/.zshrc

(mohana@kali)~[~/fabric-dev/fabric-samples/test-network]
$ source ~/.zshrc
```

7. Verify Fabric binaries

Command(s):

- peer version
- orderer version

```

$ peer version
peer:
Version: v2.5.16
Commit SHA: f871cf9
Go version: go1.26.4
OS/Arch: linux/amd64
Chaincode:
Base Docker Label: org.hyperledger.fabric
Docker Namespace: hyperledger

```

```

(mohana@kali) - [~/fabric-dev]
$ orderer version
orderer:
Version: v2.5.16
Commit SHA: f871cf9
Go version: go1.26.4
OS/Arch: linux/amd64

```

8. Navigate to the test network

Command(s):

- `cd ~/fabric-dev/fabric-samples/test-network`

9. Start the Fabric network

Command(s):

- `./network.sh up`

```

$ ./network.sh up
Using docker and docker-compose
Starting nodes with CLI timeout of '3' tries and CLI delay of '3' seconds and using database 'leveldb'
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
[+] Running 3/3
  ✓ Container orderer.example.com Running
  ✓ Container peer0.org2.example.com Running
  ✓ Container peer0.org1.example.com Running

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	NA
342344c842df	hyperledger/fabric-orderer:latest	"orderer"	About a minute ago	Up About a minute	NA
cp, [::]:7050→7050/tcp, 0.0.0.0:7053→7053/tcp, [::]:7053→7053/tcp, 0.0.0.0:9443→9443/tcp, [::]:9443→9443/tcp					or
26977a7811dd	hyperledger/fabric-peer:latest	"peer node start"	About a minute ago	Up About a minute	pe
cp, [::]:9051→9051/tcp, 7051/tcp, 0.0.0.0:9445→9445/tcp, [::]:9445→9445/tcp					pe
1000dd744ae1	hyperledger/fabric-peer:latest	"peer node start"	About a minute ago	Up About a minute	pe
cp, [::]:7051→7051/tcp, 0.0.0.0:9444→9444/tcp, [::]:9444→9444/tcp					ag
a100a25024e7	hello-world	"/hello"	10 minutes ago	Exited (0) 10 minutes ago	ag
518040179423	hello-world	"/hello"	41 hours ago	Exited (0) 41 hours ago	ag

10. Verify running containers

Command(s):

- `docker ps`

```

$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
00de63516518   hyperledger/fabric-orderer:latest   "orderer"               2 minutes ago Up 2 minutes  0.0.0.0:7050-
>7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->944
3/tcp orderer.example.com
19f90827884f   hyperledger/fabric-peer:latest     "peer node start"       2 minutes ago Up 2 minutes  0.0.0.0:7051-
>7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
peer0.org1.example.com
43090b8bd7945   hyperledger/fabric-peer:latest     "peer node start"       2 minutes ago Up 2 minutes  0.0.0.0:9051-
>9051/tcp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
peer0.org2.example.com

```

11. Create the application channel

Command(s):

- ./network.sh createChannel -c mychannel

```

$ ./network.sh createChannel -c mychannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
[+] Running 3/3
  ✓ Container orderer.example.com    Running
  ✓ Container peer0.org2.example.com  Running
  ✓ Container peer0.org1.example.com  Running
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
342344c842df   hyperledger/fabric-orderer:latest   "orderer"               About a minute ago Up About a minute
cp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
16977a7811dd   hyperledger/fabric-peer:latest     "peer node start"       About a minute ago Up About a minute
cp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
1000dd744ae1   hyperledger/fabric-peer:latest     "peer node start"       About a minute ago Up About a minute
cp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
a106a25024e7   hello-world                         "/hello"                 17 minutes ago   Exited (0) 17 minutes ago
518b40179423   hello-world                         "/hello"                 41 hours ago     Exited (0) 41 hours ago

Using docker and docker-compose
Generating channel genesis block 'mychannel.block'
Using organization 1
/home/mohana/fabric-dev/fabric-samples/test-network/.../bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mychannel.block -channelID mychannel
2026-07-24 16:10:12.342 IST 0001 INFO [common.tools.configtxgen] main -> Loading configuration
2026-07-24 16:10:12.350 IST 0002 INFO [common.tools.configtxgen.localconfig] completeInitialization -> orderer type
2026-07-24 16:10:12.350 IST 0003 INFO [common.tools.configtxgen.localconfig] completeInitialization -> Orderer.Etc
ng to tick interval:"500ms" election tick:10 heartbeat tick:1 max inflight blocks:5 snapshot interval size:16777216

```

```

+ . scripts/orderer.sh mychannel
+ '[' 0 -eq 1 ']'
+ res=0
Status: 201
{
    "name": "mychannel",
    "url": "/participation/v1/channels/mychannel",
    "consensusRelation": "consenter",
    "status": "active",
    "height": 1
}

Channel 'mychannel' created
Joining org1 peer to the channel...
Using organization 1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=0
2026-07-24 16:10:18.560 IST 0001 INFO [channelCmd] InitCmdFactory → End
2026-07-24 16:10:18.660 IST 0002 INFO [channelCmd] executeJoin → Success
Joining org2 peer to the channel...

```

12. Stop the network

Command(s):

- ./network.sh down

```

Using docker and docker-compose
Stopping network
[+] Running 7/7
✓ Container orderer.example.com      Removed
✓ Container peer0.org1.example.com    Removed
✓ Container peer0.org2.example.com    Removed
✓ Volume compose_peer0.org1.example.com Removed
✓ Volume compose_orderer.example.com  Removed
✓ Network fabric_test                 Removed
✓ Volume compose_peer0.org2.example.com Removed

```

Result

The Hyperledger Fabric development environment was successfully configured on Linux. The Fabric test network was started successfully, the application channel 'mychannel' was created, and the peer nodes joined the channel successfully.